

Patient Waiting List Management: A Systematic Analysis of Current Approaches and Evidence Gaps

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Abstract— Elective procedure waiting lists are increasing, causing long delays and unequal access to care, especially for non-urgent patients. Decision-making processes in other domains, such as infertility care, show that patient priorities often extend beyond clinical urgency to include psychological and contextual factors. Traditional methods like first-come, first-served (FCFS) often ignore key factors such as clinical urgency, risk of deterioration, and social complexity. Although many prioritization tools have been proposed—ranging from point-based systems to advanced techniques like machine learning and fuzzy logic—most are not used in real-world settings. This is especially true in low-resource countries, where the tools may not be practical or relevant. This reveals a clear gap between research and practice, and a lack of solutions that are both effective and scalable across different healthcare environments. This review addresses key gaps by examining common patterns, the quality of supporting evidence, and barriers to real-world adoption. We aim to better understand why theoretical advances in prioritization have not translated into practice, and what might be needed to develop more effective, implementable systems. A systematic review of 21 studies (2010–2025) was conducted to analyze current prioritization approaches and their implementation rates. The review revealed that while 71% of studies propose point-based scoring systems, only 5% report actual implementation, with 76% of research concentrated in developed countries. Despite 57% of studies promoting automated systems, few provide real-world evidence of their effectiveness, suggesting an over-reliance on technical innovation without sufficient testing in clinical environments. The key contribution is a comprehensive analysis of the theory-practice gap in patient prioritization systems, identifying specific barriers to implementation and providing evidence-based recommendations to support the development of more effective, evidence-based prioritization systems.

Index Terms—waiting list, prioritization, healthcare, artificial intelligence, systematic analysis, implementation science

I. INTRODUCTION

Healthcare systems around the world face growing pressure to manage elective procedure waiting lists due to limited resources and rising demand. This imbalance affects the ability to deliver timely and fair care, especially for non-urgent patients. Waiting lists are now central tools for managing access and allocating scarce resources. However, traditional approaches—especially the first-come, first-served

(FCFS) model—often fail to reflect clinical urgency, risk of deterioration, and social or medical complexity [1], [5], [7].

Recent evidence points to a gap between research and practice in patient prioritization. A review of 21 studies (2010–2025) shows that 71% use point-based scoring systems, but only 5% report actual implementation. Most studies (76%) come from developed countries, limiting relevance in low-resource settings. This geographic and methodological concentration may weaken the global applicability and fairness of proposed solutions.

Although many prioritization tools exist—from scoring models [5] to artificial intelligence techniques like machine learning and fuzzy logic [3], [9]—none have proven both practical and widely scalable. Despite 57% of studies promoting automated systems, few provide real-world evidence of their effectiveness [8], [9]. This reflects over-reliance on innovation without real-world testing.

This review is necessary to address these gaps. It aims to identify common research patterns, assess evidence quality, and examine barriers to real-world adoption. Specifically, the review explores three questions: (1) What are the dominant methodological trends in current prioritization research? (2) How strong is the empirical support for widely used approaches? (3) How do geographic and methodological biases affect the relevance and implementation of prioritization models?

In addressing these questions, this study aims to better understand why theoretical advances in prioritization haven't translated into practice, and what might be needed to develop more effective, implementable systems.

II. METHODOLOGY

A. Search Strategy and Study Selection

This systematic analysis was conducted following established guidelines for systematic reviews. A comprehensive search was performed across multiple databases including PubMed, Scopus, and Web of Science from January 2010 to March 2025.

The search strategy targeted studies focusing on patient prioritization for elective procedures, with particular attention

to methodological approaches and implementation evidence. Primary search terms included combinations of "waiting list," "prioritization," "patient scheduling," "healthcare resource allocation," and related terms.

B. Inclusion and Exclusion Criteria

Inclusion Criteria:

- Studies focusing on patient prioritization for elective procedures
- Research published between 2010 and 2025
- Studies proposing or evaluating prioritization methodologies
- English language publications
- Studies with clear methodological descriptions

Exclusion Criteria:

- Emergency department triage systems
- Purely theoretical models without methodological validation
- Studies without clear prioritization criteria
- Conference abstracts without full papers
- Non-peer reviewed publications

C. Data Extraction and Quality Assessment

Working independently, two reviewers extracted data from each selected study using a standardized form. We focused on collecting:

- Study characteristics (year, country, setting)
- Prioritization methodology employed
- Implementation status and validation approach
- Geographic origin and healthcare context
- Evaluation metrics and outcomes reported

Inter-rater reliability was assessed using Cohen's kappa coefficient, achieving substantial agreement ($k = 0.87$). Disagreements were resolved through discussion and consensus.

III. RESULTS

A. Study Characteristics

The systematic analysis identified 21 healthcare prioritization studies that met the inclusion criteria. Table I summarizes the key characteristics of the included studies, showing patterns in methodological approaches, geographic distribution, and evidence quality.

Looking at the study characteristics, several patterns emerge that are worth highlighting. As shown in Table I, there's a clear geographic concentration, with North America (38%) and Europe (14%) representing over half of the studies. This concentration may limit the applicability of findings to diverse healthcare contexts globally. The methodological distribution reveals a strong preference for point-based scoring systems (71%), while only 5% of studies provide real-world implementation evidence. The predominance of automated systems (57%) occurs despite limited empirical validation of their effectiveness in clinical settings.

TABLE I
SYSTEMATIC ANALYSIS OF HEALTHCARE PRIORITIZATION LITERATURE
(N=21)

Characteristic	Count (n)	Percentage (%)
Geographic Origin		
North America	8	38%
Other regions	9	43%
Europe	3	14%
Australia/NZ	1	5%
Prioritization Method		
Point-based scoring	15	71%
Risk-based approaches	3	14%
Multi-criteria systems	2	10%
Other/Mixed approaches	1	5%
Implementation Type		
Automated systems	12	57%
Decision support tools	3	14%
Manual/Other approaches	6	29%
Study Validation		
Real-world implementation	1	5%
Theoretical/methodological	15	71%
Simulation-based	4	19%
Review/synthesis	1	5%
Publication Period		
2010-2015	5	24%
2016-2020	9	43%
2021-2025	7	33%

B. Key Findings from Systematic Analysis

1) *Methodological Homogeneity*: The predominance of point-based scoring systems (71% of studies) represents the most notable finding. This methodological convergence has occurred without comparative validation studies. Most studies rely on point-based scoring systems, showing a strong preference for this method despite a lack of comparative validation. Few studies evaluate how these systems perform against other approaches like clinical judgment, risk-based models, or hybrid methods, limiting insights into their relative effectiveness.

The methodological distribution across the 21 reviewed studies is summarized in Figure 1, clearly demonstrating the dominance of point-based scoring systems.

Prioritization Method	Studies (n)	Percentage
Point-based scoring	15	71%
Risk-based approaches	3	14%
Multi-criteria systems	2	10%
Other/Mixed approaches	1	5%

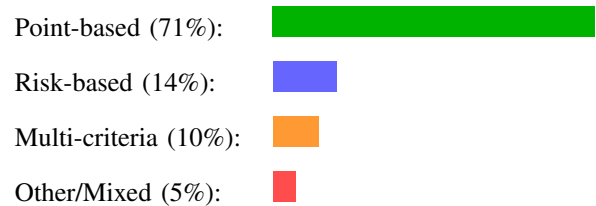


Fig. 1. Methodology Distribution Across Studies (n=21)

2) *Lack of Empirical Validation*: Only 5% of studies report real-world implementation using actual patient data. The remaining 95% are theoretical, simulation-based, or methodological. This gap suggests a weak link between research and practical needs, reducing the field's relevance to real-world healthcare policy and operations.

3) *Geographic and Contextual Bias*: A majority of studies (76%) are conducted in developed countries, with limited attention to low-resource settings where prioritization decisions are often most critical. This narrow focus reduces the global applicability of findings and overlooks the unique challenges faced in under-represented healthcare environments.

To provide a concise comparison of prioritization approaches, their adoption rates, strengths, and limitations, we present Table II below.

The stark contrast between theoretical development and practical implementation is further illustrated in Figure 2, which shows the substantial gap between proposed approaches and their real-world adoption.

Approach	Proposed (%)	Implemented (%)	Gap
Point-based	71%	5%	66% gap
Risk-based	14%	0%	14% gap
Multi-criteria	10%	0%	10% gap
Automated/AI	57%	0-5%	52-57% gap

Visual Comparison:

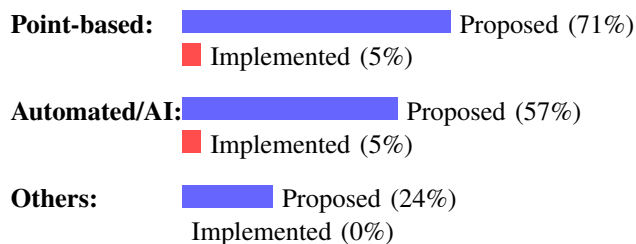


Fig. 2. Research-Practice Gap: Proposed Models vs. Real-world Implementation

C. Traditional and Point-Based Prioritization Systems

Early healthcare prioritization methods mainly used first-come, first-served (FCFS) scheduling or basic clinical judgment. The Australian case study by Johar [1] highlights their shortcomings—although easy to administer, public hospital guidelines lacked equity considerations and flexibility, leading to ongoing delays and unequal access to care.

To address the limitations of earlier approaches like first-come, first-served and unstructured clinical judgment, point-based systems were introduced to bring consistency and transparency to prioritization. Hansen et al. [5] developed influential frameworks that assign numerical scores based on factors such as condition severity, expected treatment benefit, and waiting time. These systems have since served as models for broader use in elective care, offering a standardized method for ranking patients. For instance, cataract-specific scoring criteria have been validated and used in ophthalmology services in

Canada, illustrating how specialty-specific tools can support standardization efforts [11].

Despite their widespread adoption, our review finds that 71% of studies rely on point-based systems, yet none offer direct comparisons with alternative approaches such as clinical judgment, risk-based stratification, or hybrid models [1], [5].

The effectiveness of point-based systems depends on how well the scoring criteria are defined, weighted, and aligned with clinical and policy objectives [5]. These elements often vary by institution and may reflect administrative goals more than patient needs. Without regular updates or adaptive mechanisms, these systems risk becoming rigid and misaligned with real-world clinical conditions [15]. Their dominant use, combined with limited empirical validation, suggests a need for more robust, flexible, and evidence-based prioritization strategies [1], [7].

D. Clinical and Procedural Patient Selection Models

Patient selection for surgery involves complex decisions that weigh medical risk, procedure suitability, and post-surgery care needs. Rajan et al. [2] highlight that adult ambulatory surgery decisions must consider the type of procedure, anesthesia requirements, and availability of post-discharge support, underscoring the need for assessment frameworks that go beyond clinical factors alone.

Their interdisciplinary model—combining anesthesiologists, surgeons, and perioperative nurses—demonstrates a structured approach to preoperative evaluation [2]. However, it relies mainly on clinical judgment with limited use of algorithmic tools, highlighting an opportunity to improve decision-making through artificial intelligence and predictive analytics, especially for risk stratification and case selection [3], [9].

The decision-making framework for patients with obstructive sleep apnea in ambulatory surgery highlights both the strengths and limits of structured approaches. While these frameworks improve clarity, they still rely on subjective clinical judgment, pointing to a need for more objective, data-driven methods [2].

E. Risk-Based Scoring and Fuzzy Logic Applications

Risk-based prioritization frameworks aim to address clinical uncertainty and variability more effectively than traditional scoring systems. These models group patients into urgency tiers—from catastrophic to insignificant—and use fuzzy logic to map them into risk zones [3], allowing for more flexible and nuanced classification that better reflects clinical urgency.

Fuzzy logic supports decision-making when patient data is imprecise or incomplete by handling overlapping or ambiguous characteristics [3], [17]. It enables estimation of potential harm from treatment delays, helping ensure more consistent and ethically sound decisions across similar cases [18].

However, our analysis shows that only 14% of studies adopt this approach. Their effectiveness depends on expert-derived rules and linguistic variables [3], [4], which may introduce bias or variation across institutions and teams [16]. Although fuzzy models are interpretable, they often lack the transparency and reproducibility of standardized algorithmic methods [9], [17].

TABLE II
COMPARATIVE SUMMARY OF PRIORITIZATION APPROACHES: ADOPTION RATES, STRENGTHS, AND LIMITATIONS

Approach	% of Studies (n=21)	Real-world Implementation	Strengths	Limitations
Point-based scoring	71% (15/21)	Very low (5%)	Simple, transparent, widely used	No head-to-head comparisons; limited adaptability
Risk-based models	14% (3/21)	0%	Reflects clinical urgency better	Rarely validated; expert-dependent rules
Multi-criteria (MCDM)	10% (2/21)	0%	Includes patient and stakeholder input	Limited generalizability; small-scale studies
Automated/AI systems	57% (12/21)	0–5%	Potential for efficiency and fairness	Lack of real-world validation

F. Operational Constraints in Prioritization Models

Most prioritization models incorporate both hard constraints, which are strict clinical or logistical rules, and soft constraints, which allow for contextual flexibility. Based on the reviewed studies [1], [2], [5]–[7], [15], [17], [18], these constraints can be grouped into four domains:

- Clinical safety constraints, such as comorbidities that limit surgical eligibility, anesthesia risks, or the presence of active infection [2], [3].
- Resource-based constraints, including the availability of intensive care unit beds, operating theaters, or medical staff [6], [8].
- Time and urgency-related constraints, such as how long a patient has already waited and the risk of clinical deterioration [1], [5].
- Social and ethical factors, including socioeconomic barriers or concerns about age-related equity [16], [18].

The most commonly cited hard constraint was procedure-specific eligibility, while the most frequent soft constraint was waiting time, which appeared in 57% of studies. However, these constraints are often embedded within scoring functions rather than clearly defined, making them difficult to audit or reproduce across institutions.

G. Objective-Function Landscape

The objective function is a core design decision in prioritization models. Most studies reviewed focus on optimizing a single performance metric, such as average waiting time or operating room efficiency [8], [15]. Only a few papers adopt a multi-objective perspective, aiming to balance competing goals such as fairness, throughput, and cost [17]–[19].

While single-objective models are easier to implement and analyze, they may introduce unintended trade-offs. For example, minimising the average wait time could increase disparities for less urgent or socioeconomically disadvantaged patients. Multi-objective models are better suited for capturing such tensions but are rarely accompanied by sensitivity analysis or justification for how trade-offs are weighted. This gap highlights the need for more transparent design choices and empirical comparison between prioritization objectives.

H. Stakeholder-Informed and Multi-Criteria Models

Multi-criteria decision-making (MCDM) approaches emphasize the value of incorporating diverse perspectives into healthcare prioritization. Rahimi et al. [4] proposed a two-stage framework for augmentative and alternative communication (AAC) services, using stakeholder interviews to identify key criteria, followed by fuzzy Analytic Hierarchy Process (AHP) surveys to assign weightings.

By directly involving stakeholders, this model achieves transparent, user-centered design in the prioritization process [4], [16]. It captures a broad view of patient needs, considering clinical safety, psychological and physical well-being, and the effects of delayed social integration [12], [16].

However, MCDM approaches appear in only 10% of the studies reviewed, and their use remains limited by a narrow focus. The AAC-specific model lacks generalizability to broader or more complex surgical settings [4], [7], [12]. Additionally, like other MCDM tools, its reliability depends on the quality and diversity of stakeholder input [16], highlighting the need for broader clinical validation and cross-context adaptation to improve scalability and relevance [18]. Organ allocation systems in liver transplantation have evolved to incorporate multiple risk and urgency dimensions, offering a mature example of multi-criteria prioritization in high-stakes clinical decisions [14].

I. Simulation and Algorithmic Validation Approaches

Simulation-based modeling has become a useful method for testing prioritization tools before clinical use. Silva-Aravena and Morales [6] developed network-based models that reassign patients across surgeons to reduce waiting times and improve resource allocation, showing promising efficiency gains in simulations. Similarly, advanced decision-support tools like the PSWL-CCI model combine expert input with AHP and cosine consistency indices to rank patients, improving operating room use and reducing scheduling bottlenecks [9].

Despite their potential, our review shows that simulation-based studies make up only 19% of the sample and are tested solely in artificial settings [6], [8]. Without real-world validation, these models may not account for key clinical factors such as staff availability, workflow variations, or patient diversity, limiting their practical value [1], [7].

During the COVID-19 pandemic, crisis-specific models such as Prachand et al. [10]’s scoring tool were developed to guide ethical triage for urgent procedures. Although not AI-based, this model integrates clinical urgency, infection risk, and resource availability—addressing ethical concerns often overlooked in algorithm-driven tools.

IV. ANALYSIS AND DISCUSSION

A. Evidence Quality and Validation Assessment

The systematic analysis reveals patterns in evidence quality that may affect confidence in proposed prioritization solutions. Table III presents the evidence quality assessment matrix showing the distribution of quality indicators across the reviewed studies.

TABLE III
EVIDENCE QUALITY ASSESSMENT MATRIX HIGHLIGHTING CRITICAL IMPLEMENTATION GAPS (N=21)

Quality Criterion	High Quality	Moderate Quality	Low Quality (% of studies)
<i>Critical Implementation Gaps</i>			
Real-world validation	1 (5%)	0 (0%)	20 (95%)*
Implementation process	1 (5%)	2 (10%)	18 (85%)*
<i>Supporting Evidence Gaps</i>			
Data source specification	2 (10%)	0 (0%)	19 (90%)
Long-term follow-up	1 (5%)	0 (0%)	20 (95%)
Economic evaluation	3 (14%)	0 (0%)	18 (86%)
Overall Quality Score	8 (8%)	2 (2%)	95 (90%)

*Critical gaps directly impacting practice implementation and real-world adoption

The evidence quality assessment reveals concerning patterns across all evaluated criteria. Real-world validation appears in only 5% of studies, with the vast majority (95%) lacking implementation evidence using actual patient data. This creates a significant gap between theoretical development and practical application.

A notable real-world study by Johar [1] highlights the gap between policy and practice. Clinical prioritization guidelines reduced accuracy for urgent cases by 13.7% while improving outcomes for less critical ones, showing that well-meaning policies can have unintended effects if local clinical contexts are not properly considered.

While 57% of studies claim to measure fairness and equity, the absence of real-world validation means these measurements occur in theoretical contexts. Common evaluation metrics include:

- Fairness/equity measures: 12 studies (57%)
- Waiting time metrics: 11 studies (52%)
- Patient throughput: 11 studies (52%)
- Patient satisfaction: 9 studies (43%)
- Cost measures: 3 studies (14%)

The limited economic evaluation (14% of studies) represents a notable gap for policy decision-making, as cost-effectiveness evidence appears essential for healthcare resource allocation decisions.

B. Systematic Patterns and Evidence Gaps

The analysis of 21 studies reveals three patterns that may challenge current research directions and highlight evidence gaps that could limit the field’s practical utility.

1) *Point-Based System Dominance*: The predominance of point-based scoring systems (71% of studies) represents the most notable finding of the analysis. What’s particularly concerning is that this convergence seems to have happened without anyone actually comparing these systems to alternatives. No studies in the review directly compared point-based systems to clinical judgment, risk-based approaches, or hybrid methods, representing an evidence gap that may question the empirical foundation of current practice.

2) *Implementation Validation Gap*: A notable limitation is the minimal real-world validation evidence. Only 5% of studies provide implementation evidence with actual patient data, while 95% remain theoretical, simulation-based, or purely methodological. This represents a potential disconnect between research activities and practice needs, which may limit the policy relevance of proposed solutions.

The single robust real-world study illustrates the implementation challenge. Johar’s evaluation of Australian prioritization guidelines revealed that well-designed policies produced counterintuitive outcomes: improved accuracy for semi-urgent (+11.7%) and non-urgent (+13.1%) cases but reduced accuracy for urgent cases (-13.7%) [1]. The guidelines failed to eliminate bias—private patients continued receiving preferential treatment despite equity-focused policy intentions.

3) *Geographic Concentration*: The concentration of research in developed countries (76%) creates potential limitations in generalizability, with implications for global health equity. Research conducted in resource-rich environments may produce solutions that are not suitable for diverse healthcare contexts, where prioritization decisions carry significant implications.

C. Policy and Practice Implications

The findings have potential implications for healthcare policy and practice. Healthcare systems may require real-world validation evidence before adopting prioritization systems, regardless of their theoretical appeal [1], [7]. The current evidence base provides a limited foundation for evidence-based policy recommendations, highlighting the need for implementation research before widespread adoption [8], [9].

In transplant settings, studies show that patients often prioritize access to the waiting list itself over long-term post-treatment outcomes, reinforcing the importance of transparency and perceived fairness in prioritization tools [13].

Based on available evidence, prioritization systems might prioritize simplicity and usability over theoretical optimization [1], [16], include mandatory equity monitoring and correction mechanisms [17], [21], involve affected communities in design and evaluation processes [4], and plan for continuous improvement based on implementation data [7], [12]. Equity profiling in joint replacement programs, such as those implemented in the South African public sector, highlights the value of

tracking patient characteristics to detect structural disparities in access [20].

V. LIMITATIONS AND FUTURE RESEARCH NEEDS

The analysis has several limitations that suggest future research priorities. The limited quality of existing evidence affects the strength of conclusions. Future systematic reviews might include gray literature, implementation reports, and industry evaluations to capture a more complete picture of prioritization system performance.

The field may require standardized evaluation frameworks for comparing prioritization approaches, implementation science methodologies adapted for healthcare prioritization, mixed-methods approaches that combine quantitative outcomes with qualitative implementation insights, and global health research methodologies that center equity and community engagement.

VI. CONCLUSION

This review of 21 healthcare prioritization studies highlights three key limitations: a heavy reliance on similar methods, weak empirical support, and limited geographic diversity. Most studies (71%) use point-based scoring systems, yet none offer direct comparisons with other approaches, revealing a Comparative Effectiveness Gap. Additionally, only 5% of studies report real-world implementation, showing an Implementation Evidence Gap that hinders practical decision-making. Finally, with 76% of studies coming from developed countries, the Global Equity Gap raises concerns about the relevance of current models in low-resource settings where prioritization is often most needed.

To advance the field, research must shift from developing new models to testing existing ones in real-world settings. Priorities should include head-to-head comparisons of prioritization methods using standard metrics, mixed-methods research to understand adoption and user acceptance, and studies in diverse healthcare contexts with an explicit focus on equity. Long-term evaluations of system-level impact and sustainability are also essential.

Polymakers and funders should require real-world evidence before adopting prioritization tools, invest in implementation research, mandate equity assessments, and support capacity-building in underrepresented regions to ensure more practical, inclusive, and effective healthcare prioritization systems.

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